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|---|---|---|
| **6(a)** | A line emission spectrum shows that only certain values of wavelengths (or frequencies) of photons are emitted when gas atoms de-excite.<br><br>Photons emitted have energies equal to the differences between the pairs of electron energy levels involved in the de-excitation.<br><br>The fact that only certain wavelengths are seen implies that only photons of certain energies are emitted, which in turn implies that electrons inside an atom can only possess certain discrete values of energies (called the electron energy levels). | **B1**<br><br>**B1**<br><br>**B1** |
| **6(b)(i)** | Use ruler to measure<br><br>n = 2 is - 3.4 eV (accept values between -3.3 to -3.5 eV)<br>n = 3 is -1.5 eV (accept values between -1.4 to -1.6 eV) | <br><br>**A1**<br>**A1** |
| **6(b)(ii)** | Energy of photon = (-1.5 – (-3.4)) eV<br><br>$\frac{hc}{\lambda}$ = (-1.5 – (-3.4)) (1.6 x $10^{-19}$)<br>$\frac{6.63x10^{-34}x3x10^{8}}{\lambda}$ = (-1.5 – (-3.4)) (1.6 x $10^{-19}$)<br><br>$\lambda$ = 6.54 x $10^{-7}$ m (For info: visible region) | <br><br>**C1**<br><br>**A1** |
| **6(b)(iii)** | Accept any of the following:<br>1. By thermal excitation by heating up the hydrogen gas. (electrons of high energy)<br>2. By applying high voltage across the gas using a voltage discharge tube/ cathode ray tube<br>3. Shine white light on the gas (use photons of all wavelengths)/ expose to sunlight<br>4. Shine light of frequency / wavelength / photon energy corresponding to energy difference of energy levels | **B1** |